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Original Article

Decreased pH and increased temperatures affect young-of-the-year red king crab (*Paralithodes camtschaticus*)

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The red king crab (*Paralithodes camtschaticus*) is a high-latitude commercially important species with a complex life-history cycle which encompasses a wide variety of conditions and habitats. High-latitude waters, including those around Alaska where red king crab live, are predicted to have increased ocean acidification and temperatures in comparison to other areas. The interaction of ocean acidification and increased temperature has not been examined for any life history stage of red king crab. To determine the effects of near-future ocean acidification and warming temperature on young-of-the-year red king crab survival, growth, and morphology, we conducted a long-term (184 d) fully crossed experiment with two pHs and three temperatures: ambient pH (~7.99), pH 7.8, ambient temperature, ambient +2 °C, and ambient +4 °C, for a total of six treatments. Mortality increased with exposure to reduced pH and higher temperatures, but a clear trend in the interactive effects of the stressors was not observed. A synergetic effect on mortality was observed in the pH 7.8 and ambient +4 °C temperature treatment. This treatment also had the lowest survival with only 3% surviving to the end of the experiment. However, an antagonistic effect on mortality was observed in the pH 7.8 and ambient +2 °C treatment. Lower pH and warmer temperatures affected intermoult duration, only temperature affected percent increase in size, but carapace length was not affected. Decreased pH and increased temperature had no effect on morphology. The results of this study combined with other studies show that decreased pH and warming has profound negative effects on red king crab. Unless the species is able to adapt or acclimate to changing climate conditions, red king crabs populations may decrease in the upcoming decades due to ocean acidification and rising temperatures.

Keywords: growth, morphology, ocean acidification, ocean warming, *Paralithodes camtschaticus*, red king crab, survival, young-of-the-year.

Introduction

Ocean acidification, a reduction in oceanic pH due to uptake of anthropogenic CO_2, is predicted to increase for centuries and result in an approximate global ocean surface pH of 7.8 by the end of this century (Caldeira and Wickett, 2003; Bopp *et al.*, 2013; IPCC, 2014). In addition, the world's oceans are warming due to climate change and by the end of the century global sea surface temperatures are predicted to increase by ~3 °C (Bopp *et al.*, 2013). Ocean acidification and warming rates vary throughout the world's oceans. High-latitude waters, including the waters around Alaska, are predicted to have increased ocean acidification and temperatures in comparison to other areas (Fabry *et al.*, 2009; Mathis *et al.*, 2015). Colder waters hold more CO_2 than warmer waters and thus have lower calcium carbonate saturation levels than warmer waters (The Royal Society, 2005; Takahashi *et al.*, 2014). From pre-industrial times to the present (2003–2012), the surface pH in the Bering Sea and Gulf of Alaska decreased by 0.10 units (Mathis *et al.*, 2015), which is the same as the average global estimate (Caldeira and Wickett, 2003; IPCC, 2014). However, from the present to the year 2095, surface pH reduction and warming is projected to be 0.35 pH units and 4.15 °C in the Bering Sea and 0.34 pH units and 3.40 °C in the Gulf of Alaska (Mathis *et al.*, 2015), which are higher than the global estimated changes expected over the same period (Bopp *et al.*, 2013;

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IPCC, 2014). By the end of the century, all waters around Alaska are predicted to be perennially undersaturated with regard to aragonite, and only the Bering Sea and Gulf of Alaska will remain supersaturated with respect to calcite (Mathis *et al.*, 2015).

Multiple stressor studies are needed to better understand the effects of oceanic changes (e.g. ocean acidification, warming water, decreased oxygen, rising sea levels, etc.) on marine organisms (IPCC, 2014; Breitburg *et al.*, 2015; Gunderson *et al.*, 2016). Multiple stressor effects can be additive, antagonistic (less than the sum of individual effects), or synergistic (greater than the sum of individual effects). The intensity and timing of stressors often impact the response. Additive effects are expected if multiple stressors are introduced relative far apart in time, antagonistic effects are likely when stressors are introduced asynchronously and over relatively short-time periods, and synergistic effects are likely when the stressors are introduced simultaneously (Gunderson *et al.*, 2016). A meta-analysis of ocean acidification and warming temperature studies reports a trend of increased sensitivity to ocean acidification with increased temperature (Kroeker *et al.*, 2013); however, the responses vary by study, taxon, and life history stage. Byrne and Przeslawski (2013) report additive or antagonistic effects as being the most common responses, with few cases of synergistic effects, but Harvey *et al.* (2013) report synergistic effects as the most common response.

The red king crab, *Paralithodes camtschaticus*, is a high latitude, commercially important species found in the North Pacific Ocean and the Barents Sea (where they were introduced) (Donaldson and Byersdorfer, 2005). They have a complex life-history cycle which encompasses a wide variety of conditions and habitats. Females brood eggs for approximately a year, depending on their life-history stage; both the brooding cycle and the size-fecundity relationship vary between primiparous (brooding first clutch) and multiparous (brooding second or subsequent egg clutch) crabs (Stevens and Swiney, 2007; Swiney and Long, 2015). Larvae are released in the spring and spend about 2–3 months in the plankton, where they pass through four zoeal stages and one glaucothoe (settling) stage (Shirley and Shirley, 1989). Glaucothoe settle when they find complex habitats such as hydroids or cobble (Loher and Armstrong, 2000; Stevens, 2003), because predation is reduced in such habitats (Stevens and Swiney, 2005; Long and Whitefleet-Smith, 2013) both by conspecifics (Long *et al.*, 2015) and other species (Stoner, 2009; Pirtle *et al.*, 2012). Such habitats are typically found in near-shore areas (McMurray *et al.*, 1984), and red king crab juveniles are found from the intertidal down into the subtidal (Wainwright *et al.*, 1992). By about the age of 2 years, juveniles have become too large to hide in complex habitats; they form pods (aggregations), and move into deeper, unstructured habitats (Dew, 1990), where they typically remain as they mature to the adult stage.

Because of the ontogenetic changes in habitat requirements in red king crab, and because those habitats have different temperatures and temperature variability, it is not surprising that different life-history stages of red king crab vary in their thermal tolerance. Embryos are the most sensitive to higher temperatures, with mortality increasing at temperatures above 8 °C (Nakanishi, 1987). Mature females are less sensitive to higher temperatures, but they still experience sub-lethal effects of reduced growth at 12 °C (Shirley *et al.*, 1989). Mature females, and therefore embryos, are generally distributed in deeper, colder habitats with more stable temperatures. On the other hand, larvae, which exist in the upper water column, and juveniles, which are distributed as shallow as the intertidal, both have higher thermal tolerances. Larvae culturing is effective at 11 °C in the lab (Swingle *et al.*, 2013), and increased mortality occurs at 15 °C (Kurata, 1960). Juveniles have a similar thermal tolerance, with no effects at 12 °C and sublethal effects of decreased growth occurring at 15 °C along with a slight increase in mortality during moulting (Rice *et al.*, 1985; Stoner *et al.*, 2010); significant acute mortality occurs at 24 °C (W. C. Long, unpublished data).

Although the effects of ocean acidification on red king crab have been less studied than the effects of temperature, there is good evidence that red king crab are sensitive to ocean acidification. At a pH of 7.7, embryos and larvae both had a change in morphology and larvae had a decrease in starvation survival time (Long *et al.*, 2013a). Juveniles are also sensitive, with increased mortality and decreased growth at pH 7.8, and very high mortality at pH 7.5 (Long *et al.*, 2013b). The effects are such that their populations, and the fisheries that depend on them, are likely to be negatively affected by the end of the century (Punt *et al.*, 2014).

The interaction of ocean acidification and increased temperatures has not been examined for any life history stage of red king crab, and given that the two stressors are likely to co-occur, it is critical to understand their combined effects. To determine the effects of near-future ocean acidification and warming temperature on young-of-the-year (juveniles that are <1-year old) red king crab survival, growth, and morphology, we conducted a long-term, fully crossed experiment with two pHs and three temperatures: ambient pH (~7.99), pH 7.8, ambient temperature, ambient +2°C, and ambient +4°C, for a total of 6 treatments. A pH treatment lower than pH 7.8 was not included in this study because in a previous study 100% mortality occurred within 95 d for young-of-the-year red king crab exposed to pH 7.5 waters (Long *et al.*, 2013b).

Methods

Seawater acidification and heating

Seawater was acidified using the same methods described in Long *et al.* (2013a). Briefly, sand filtered seawater was pumped into the Alaska Fisheries Science Center's Kodiak Laboratory seawater facility from 15 and 26 m depth intakes in Trident Basin. A tank of pH 5.5 water was established by bubbling CO_2 into ambient seawater. This pH 5.5 water was mixed with ambient seawater in the treatment head tanks via peristaltic pumps controlled by Honeywell controllers and Durafet III pH probes. The ambient head tank did not receive any pH 5.5 water. Waters from the treatment head tanks were then supplied to the treatment tubs (one tub per treatment). To heat the water, a 200W submersible heater was placed in each heated treatment tub. In the coldest months of the experiment, a 100W heater was added to the warmest treatments to maintain the correct temperatures. Temperature and pH_F (free scale) were measured daily from three random inserts in each treatment tub (see below for experimental set-up details) using a Durafet III pH probe calibrated with a TRIS buffer, and when the pH deviated from the target pH by more than ±0.02 pH units, the Honeywell controller set points were adjusted to bring the pH back to the target value. Heater set points were changed manually based upon ambient temperature for each day to maintain target treatment temperatures. Weekly water samples from the treatment head tanks were taken, fixed with mercuric chloride, and sent to analytic laboratories for dissolved inorganic carbon (DIC) and total alkalinity (TA) analysis. The two laboratories used similar, but slightly different,

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instruments (CM5014 Coulometer with a CM5130 Acidification Module (UIC Inc., Joliet, IL) and VINDTA 3C (Marianda, Kiel, Germany) coupled to a 5012 Coulometer (UIC Inc., Joliet, IL)). Both laboratories used Certified Reference Material from the Dickson Laboratory (Scripps Institution, San Diego, CA).

Laboratory study

Ethical approval for this research was not required by any federal, state, or international laws because the study was conducted on invertebrates which are not covered under these laws.

Young-of-the-year crab were reared in the Kodiak Laboratory from an ovigerous female collected in Bristol Bay, Alaska, June 2011 and shipped live to the laboratory. Thirty young-of-the-year crab were randomly assigned to each of six treatments: (i) ambient pH, ambient temperature; (ii) ambient pH, ambient +2°C; (iii) ambient pH, ambient +4°C; (iv) pH 7.8, ambient temperature; (v) pH 7.8, ambient +2°C; and (vi) pH 7.8, ambient +4°C. Each treatment was contained in a 53 (L) × 38 (W) × 23 (H) cm tub. Each tub had a flow rate of 600 ml min^{-1} from the head tank. The young-of-the-year crab were reared in individual inserts constructed from 40 mm inner diameter PVC pipe with 750 μm mesh attached to the bottom and the inserts were placed inside the treatment tub. The area of this insert was determined to be optimal for individual rearing of juvenile red king crab of the size used in this study (Swiney *et al.*, 2013b). Inserts were placed on plastic grating to raise the inserts off the bottom of the tub. Water was delivered into each insert from above via a submersible pump connected to a manifold which recirculated water within each tub. A temperature logger that recorded data every 30 min was placed into each tub. To acclimate the crab to the treatment temperatures, each day the temperature in the tubs was increased by 1°C until the desired temperature was obtained. Day 1 of the experiment (5 August 2012) was marked when all of the treatments reached the correct temperature; the experiment ran for 184 d.

Target temperatures were achieved throughout the experiment, except for 8 August 2012, when a manifold hose came off in the ambient pH, ambient temperature treatment and the temperature spiked in that tub (Figure 1). Likewise, target pHs were achieved throughout the experiment, but because the individual treatment tubs were heated, pHs were slightly lower in the warmer treatments (Table 1). The DIC, bicarbonate and carbonate differed significantly between the pH treatments, but did not differ among the temperature treatments within each pH treatment (Table 1). The pCO_2 increased with decreasing pH, and alkalinity did not vary with treatment. Aragonite was supersaturated in the ambient treatment, but undersaturated in the pH 7.8 treatment; calcite was supersaturated in both treatments (Table 1).

Throughout the duration of the experiment, crab were fed a gel diet of Gelly Belly (Florida Aqua Farms, Inc., Dade City, FL, USA) enhanced with Cyclop-eeze powder (Argent Laboratories, Redmond, WA, USA), pollock bone powder (US Department of Agriculture, Agricultural Research Service, Kodiak, AK, USA), and astaxanthin (Daly *et al.*, 2012). Crab were fed to excess three times per week and uneaten food was removed prior to each feeding. Each insert was checked daily for exuvia and mortalities which were removed for growth and morphometric analysis. Carapaces were carefully removed from all exuvia and mortalities and photographed under a dissecting microscope. Carapace width (CW), carapace length (CL), rostrum base width (RW), orbital spine width (OW), and the first spine length (SL) (Figure 2) were measured in mm using Image Pro Plus v. 6.00.260 imaging software (Media Cybernetics, Inc., Bethesda, MD, USA, Long *et al.*, 2013b).

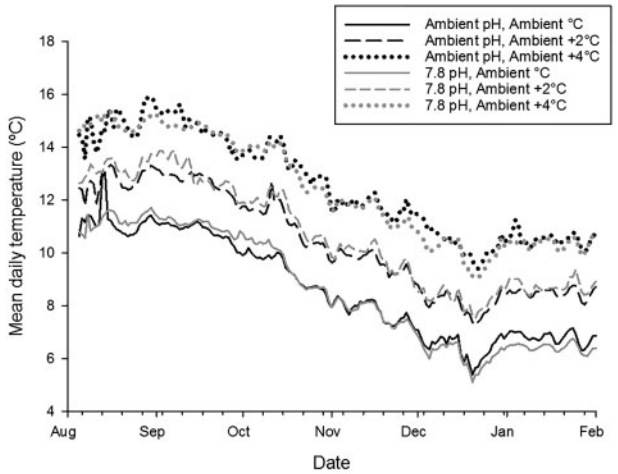

Figure 1. Mean daily temperatures (°C) for each treatment for the duration of the experiment.

Statistical analysis

We assumed a constant mortality rate and modelled mortality as an exponential loss function such that:

$$N_t = N_I e^{-mt}$$

where N_t is the number of crabs alive at time t, N_I is the initial number, m is the morality rate, and t is the time in days. We fit the number of crabs alive in each treatment to a series of models where m was a linear function of the pH and temperature treatments in R 2.14.0 (Vienna, Austria) assuming a binomial distribution using maximum likelihood. We calculated the AIC_c (Akaike's Information Criterion corrected for sample size) for each model and used it to select the most parsimonious model; models with AIC_cs that differed by <2 were considered to explain the data equally well (Burnham and Anderson, 2002).

Intermoult durations, the time between moulting events, were analysed with fully crossed general linear models using both experimental days and developmental degree days. Developmental degree days were calculated as the sum of the average daily temperatures. Experimental or developmental degree days were the dependent variable, individuals were nested, and pH, temperature, and intermoult number (first or second intermoult duration) were the factors. Intermoult duration could not be calculated between the beginning of the experiment and the first moult because it was impossible to determine where crabs were in their moulting cycle at the beginning of the experiment. Growth, measured as a change in CL, was analysed with fully crossed general linear models for mean initial size, size after each moult, and size at the end of the experiment. Percent increase in size between moults was also analysed with a fully crossed general linear model. CL or percent increase in size were the dependent variables, and pH, temperature, and period of growth for only the percent increase analysis were the factors. Individuals were nested in the percent increase in size model. When statistically

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Table 1. The mean and s.d. of water chemistry parameters from the treatment head tanks.

Treatment	pH_F	pCO_2 µatm	HCO_3^- mmol/kg	CO_3^{-2} mmol/kg	DIC mmol/kg	Alkalinity mmol/kg	$\Omega_{Aragonite}$	$\Omega_{Calcite}$
Ambient pH,	7.99^A	491.28^A	1.90^A	0.09^A	2.01^A	2.12^A	1.34^A	2.13^A
Ambient °C	(0.03)	(40.04)	(0.07)	(0.01)	(0.07)	(0.07)	(0.13)	(0.20)
Ambient pH,	7.96^B	538.00^B	1.90^A	0.09^A	2.01^A	2.12^A	1.33^A	2.11^A
Ambient +2 °C	(0.03)	(41.22)	(0.07)	(0.01)	(0.07)	(0.07)	(0.12)	(0.19)
Ambient pH,	7.94^C	587.97^C	1.90^A	0.09^A	2.01^A	2.12^A	1.35^A	2.12^A
Ambient +4 °C	(0.03)	(40.94)	(0.07)	(0.01)	(0.07)	(0.07)	(0.12)	(0.19)
pH 7.8, Ambient °C	7.79^D (0.02)	826.03^D (24.29)	1.97^B (0.05)	0.06^B (0.004)	2.07^B (0.05)	2.11^A (0.05)	0.86^B (0.06)	1.36^B (0.09)
pH 7.8, Ambient +2 °C	7.76^E (0.02)	896.11^E (26.58)	1.97^B (0.05)	0.06^B (0.004)	2.07^B (0.05)	2.11^A (0.05)	0.87^B (0.06)	1.37^B (0.10)
pH 7.8, Ambient +4 °C	7.74^F (0.01)	955.08^F (32.81)	1.97^B (0.05)	0.06^B (0.004)	2.07^B (0.05)	2.11^A (0.05)	0.89^B (0.07)	1.40^B (0.10)

The pH_F (free scale) was measured daily ($n = 184$), DIC and alkalinity were measured weekly ($n = 26$), and all other parameters were calculated. Ω is the saturation state. Different letters indicate significant difference for each parameter among treatments (ANOVA, $p < 0.05$).

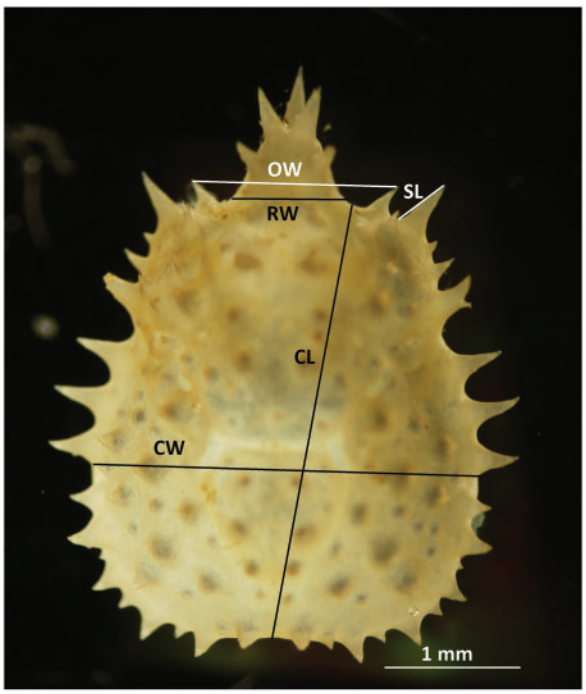

Figure 2. Young-of-the-year red king crab morphometric measurements: CW, CL, RW, OW, and the first SL.

significant differences were detected in response variables, Tukey's *post-hoc* multiple comparison tests were used to examine differences among treatments. The number of moulting events varied with treatment (Table 2); thus, intermoult duration and growth indices were only analysed when data had been acquired for all the treatments. The assumption of homogeneity of variance was tested using Levene's test and all data met this assumption.

Red king crab morphology was visualized using non-metric multidimensional scaling (n-MDS) by means of a Bray-Curtis resemblance matrix. Differences among treatments were analysed with fully crossed, three-way permutational, multivariate analysis of variance (PERMANOVA), with pH treatment, temperature treatment, and moult stage as factors preceded by a permutation dispersion analysis (PERMDISP) to check the assumption of homogeneity of multivariate dispersion. Multivariate statistics were performed in Primer 6.1.15 with PERMANOVA+ 1.0.5 (Plymouth, UK).

Results

In the best fitting model, the mortality rate varied with pH, temperature, and their interaction; no other model had any support (Table 3). The mortality rate was increased by 82% with low pH and by 49–107% with higher temperatures (Table 3). There were interactive effects; crabs held at pH 7.8 and ambient +2°C temperatures had antagonistic effects (less than additive) and those held at pH 7.8 and ambient +4°C temperatures had synergistic effects (greater than additive) (Table 3; Figure 3). At the end of the experiment, the lowest survival to the highest survival was as follows: pH 7.8 and ambient +4 °C temperature treatment, ambient +2 °C temperature treatment, pH 7.8 treatment, ambient +4 °C temperature treatment, pH 7.8 and ambient +2 °C temperature treatment, and ambient pH and ambient temperature treatment (Figure 3).

Increasing temperature significantly decreased the intermoult duration by 8–22 experimental days depending upon which temperature treatments were compared (Table 4; Figure 4a). Intermoult duration was significantly longer, on average 5 experimental days, for crab in the pH 7.8 treatments compared with the ambient treatments, and the first intermoult duration was significantly shorter, 29 experimental days, than the second intermoult duration (Table 4; Figure 4a). Intermoult duration in developmental degree days did not vary with temperature (Table 4; Figure 4b). Young-of-the-year in the pH 7.8 treatments had a significant and slightly longer intermoult duration of 64 developmental degree days more than those in the ambient treatments (Table 4; Figure 4b). The first intermoult duration was significantly shorter, by 232 developmental degree days, than the second intermoult duration (Table 4, Figure 4b).

Mean CL did not vary with treatment at the beginning of the experiment, after each of the first three moults, or at the end of the experiment (Table 5). Initial CLs ranged from 2.39 to 3.77 mm, after the first moult CLs were 2.64–4.51 mm, after the second moult CLs were 3.08–4.98 mm, after the third moult CLs were 3.27–5.43 mm, and at the end of the experiment CLs were 3.02–5.19 mm. The percent increase in CL did not vary with pH, but decreased significantly in the ambient +4 °C treatment compared with the ambient temperature treatment by 2.5%, and also decreased significantly with consecutive periods of growth (Table 6; Figure 5). Mean percent increase in CL after the first moult was

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Table 2. Number of young-of-the-year red king crab that underwent each moult by treatment.

Treatment	First moult	Second moult	Third moult	Fourth moult	Fifth moult
Ambient pH, Ambient °C	30	23	14	0	0
Ambient pH, Ambient +2 °C	30	21	11	1	0
Ambient pH, Ambient +4 °C	30	26	20	8	2
pH 7.8, Ambient °C	30	26	9	1	0
pH 7.8, Ambient +2 °C	30	28	14	6	0
pH 7.8, Ambient +4 °C	29	22	5	2	0

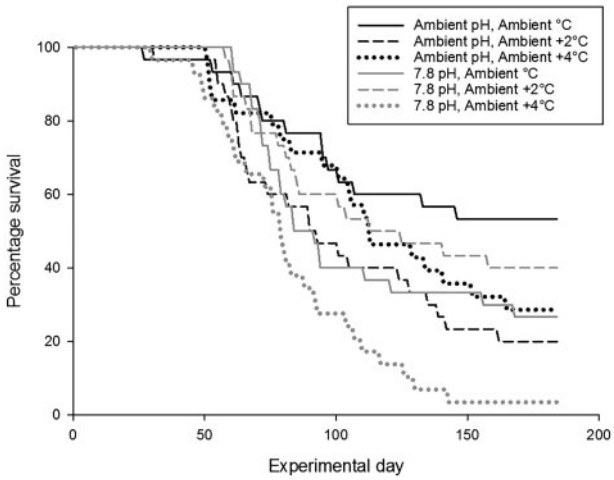

Figure 3. Percentage survival of young-of-the-year red king crab by experimental day for all treatments.

Table 3. AIC_cs for models of young-of-the-year red king crab mortality and parameter estimates for the best fit model.

Model	K	AIC_c	ΔAIC_c	Likelihood	AIC_c weight
m	1	8518.95	1620.27	0.00	0.00
m(pH)	2	8243.34	1344.66	0.00	0.00
m(T)	3	8050.23	1151.54	0.00	0.00
m(pH,T)	4	7802.21	903.52	0.00	0.00
m(pH#T)	6	6898.69	0.00	1.00	1.00
	Parameter		**Estimate ± s.e. (d^{-1})**		
	m		0.00336 ± 0.00009		
	$m_{7.8}$		0.00276 ± 0.00016		
	m_{+2}		0.00361 ± 0.00017		
	m_{+4}		0.00163 ± 0.00015		
	$m_{7.8,+2}$		−0.00522 ± 0.00025		
	$m_{7.8,+4}$		0.00346 ± 0.00031		

Model specifies whether the mortality rate *m* was a function of pH (pH), temperature (T) with # indicating a fully crossed model. K is the number of parameters. Parameters include the base mortality rate *m* and the effects (subscripted) of pH 7.8 (7.8), ambient +2 temperatures (+2), and ambient +4 temperatures (+4). Interactive effects are indicated with two treatments in the subscript. Total sample size was 1104.

18.67 (s.d. = 5.85)%, followed by 5.97 (s.d. = 4.85)% after the second moult, and 2.74 (s.d. = 4.03)% after the third moult.

Red king crab morphology did not vary with pH or temperature or with any of the interaction terms (Table 7; Figure 6). Dispersions did not vary with pH (PERMDISP, $p = 0.197$) or temperature ($p = 0.727$) treatments, but did among moult stage ($p = 0.001$). Thus, although we can state that the morphology

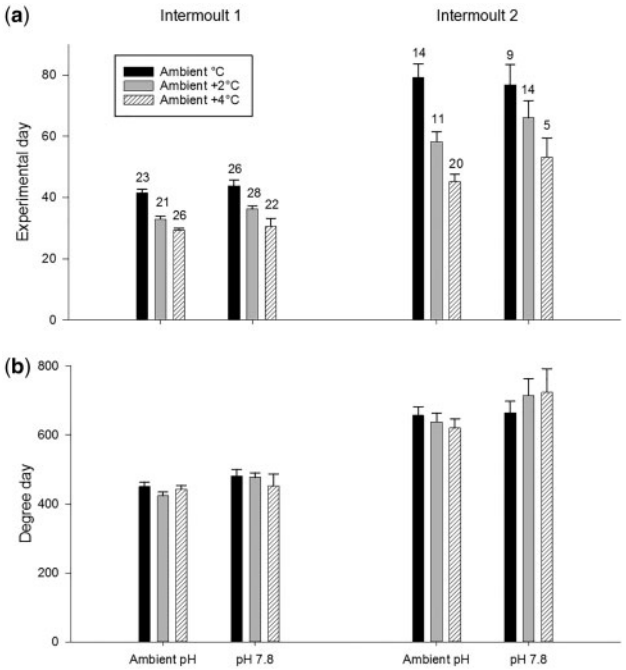

Figure 4. Mean young-of-the-year red king crab intermoult durations for all treatments. (a) intermoult duration in experimental days, (b) intermoult duration in developmental degree days. Bars are mean ± s.e. and numbers above bars are sample sizes, which are the same for experimental and degree days.

does not differ among pHs, temperatures, and the interactive terms based on the PERMANOVA (Table 7), differences in morphology among moult stages could be driven by either differences in morphology and multivariate dispersion or just differences in dispersion (Anderson *et al.*, 2008). The former is almost certainly the case here (Figure 6) as it is consistent with the growth expected from one moult stage to the next (Westphal *et al.*, 2014); as size increases (a change in morphology), the variance in size also increases (a change in dispersion).

Discussion

Red king crab young-of-the-year were sensitive to both increased temperature and reduced pH in this study, with the combination of the highest temperature and lowest pH having a synergistic effect on mortality. Lower pH and warmer temperatures affected intermoult duration, only temperature affected percent increase in size, but CL was not affected. Decreased pH and increased temperature had no effect on morphology. That 97% of young-of-the-year red king crab died at a temperature/pH combination mimicking what is expected within 80 years suggests that this

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Table 4. Fully crossed general linear models of young-of-the-year red king crab intermoult duration in experimental and developmental degree days.

	Factor	*F*-ratio	*p*-value
Experimental days	pH	8.961	0.004
	Temp	59.670	<0.001
	IM	315.379	<0.001
	pH × Temp	0.691	0.505
	pH × IM	2.532	0.116
	Temp × IM	10.789	<0.001
	pH × Temp × IM	0.707	0.497
	Crab (pH × Temp)	1.668	0.010
Degree days	pH	21.585	<0.001
	Temp	0.677	0.512
	IM	281.011	<0.001
	pH × Temp	0.640	0.530
	pH × IM	5.700	0.020
	Temp × IM	0.037	0.964
	pH × Temp × IM	1.311	0.276
	Crab (pH × Temp)	2.179	<0.001

"pH" denotes treatment pH, "Temp" denotes treatment temperature, "IM" denotes intermoult number; e.g. first or second intermoult, and "Crab" denotes individual crab that were nested in the model.

species may be significantly affected by climate change in the near future unless the species is able to quickly adapt or acclimate to changing conditions. Possible acclimation of the congener blue king crab (*Paralithodes platypus*) under acidified conditions has been observed. In pH 7.5 waters, juvenile blue king crab had an initial high rate of mortality that dropped rapidly and by the end of a year-long experiment, the mortality rate in the pH 7.5 treatment was similar or lower to the mortality rate in ambient pH and pH 7.8 treatments suggesting a high degree of phenotypic plasticity and possibly acclimation (Long *et al.*, 2017). We did not observe similar changes in mortality rate nor possible acclimation of young-of-the-year red king crab in this study.

Temperatures in the ambient and ambient +2 °C treatments should not have exceeded thermal tolerances for this life stage, but the ambient +4 °C temperature treatment may have been sufficient to cause both sublethal and lethal effects (Rice *et al.*, 1985; Jewett and Onuf, 1988; Stoner *et al.*, 2010). Further, exposure to acidified waters can reduce crab thermal tolerances (Metzger *et al.*, 2007; Walther *et al.*, 2009; Whiteley, 2011), which may have occurred in the pH 7.8 ambient +4 °C temperature treatment, resulting in the synergetic effect of increased mortality rate observed. Exposure to acidified waters can also increase thermal tolerances as was observed in the intertidal porcelain crab (*Petrolisthes cinctipes*) (Paganini *et al.*, 2014). Why the ambient pH ambient +2 °C temperature treatment had the second highest overall mortality is unclear since this treatment temperature peaked at 13 °C, which should be within the thermal tolerance for young-of-the-year red king crab. Given that, and the lower overall mortality in our +4 °C temperature treatment which appears more consistent with previous juvenile red king crab research, we suspect the mortality estimate for this treatment is high; however, the difference could also be partially due to experimental duration. Previous juvenile red king crab experiments examining the effects of temperature have only been run for 60–90 d (Rice *et al.*, 1985; Stoner *et al.*, 2010); in our experiment, virtually no mortality occurred in the first 50 d. Godbold and Solan (2013) caution against drawing conclusions from short-term experiments when examining the potential effects of ocean acidification and warming on marine organisms as it may take several months for effects to be observed.

Antagonistic effects were observed in the pH 7.8 ambient +2 °C temperature treatment since the mortality rate in this treatment was less than the mortality rate of each of the stressors individually. Antagonistic effects have been reported in acidification and temperature multistressor studies for other species (see review Byrne and Przeslawski, 2013), but to our knowledge this is the first study where antagonistic effects have been observed with crab. However, if our estimate of the effects of the ambient +2 °C temperature treatment is an overestimate, as we suspect, the magnitude of the antagonistic effect may be commensurately lower. Percent survival in the pH 7.8 ambient temperature treatment was similar to the survival observed in another young-of-the-year red king crab study at pH 7.8 at ambient temperature run for a comparable duration (Long *et al.*, 2013b), suggesting a consistency in the survival response of this life stage under these conditions.

Energy balance is important for an organism's ability to tolerate environmental stressors and some activities, such as somatic maintenance, cannot be reduced below a certain level if the organism is to survive (Sokolova *et al.*, 2012). In this study, virtually no mortality was observed in any of the treatments within approximately the first 50 d of the experiment suggesting that in the shorter term these crab were able to maintain critical processes under the stress of decreased pH and/or increased temperatures and survive, but that ability was limited and could not be maintained for a longer duration. Likewise, decreased pH did not increase mortality of juvenile porcelain crab until after 40 d (Ceballos-Osuna *et al.*, 2013). Energetic trade-offs under warming and acidified conditions have been observed for other decapods. In the short term (2.5 weeks), porcelain crab had a smaller overall aerobic energy budget and devoted a larger portion of that budget to basal maintenance under acidified and warming conditions which may lead to negative long-term fitness (Paganini *et al.*, 2014). Juvenile European lobsters (*Homarus gammarus*) had increased mortalities under acidified and warming conditions, likely due to the observed reductions in metabolism and food acquisition (Small *et al.*, 2016).

Crustacean growth rate is a function of the intermoult duration (time between moults) and percent increase in size with each moult (Hartnoll, 2001). Typically, growth rate increases with increasing temperature because the intermoult duration is shortened (Hartnoll, 2001), and red king crab growth in this study was not an exception; increased temperature resulted in decreasing intermoult duration in experimental days. Decreased pH increased young-of-the-year red king crab intermoult duration in both experimental and developmental degree days, but the effects were small. For other decapod species, decreasing pH has differing effects on intermoult duration. For example, intermoult duration increased for American lobster larvae (*Homarus americanus*), spider crab larvae (*Hyas araneus*), and juvenile southern Tanner crab (*Chionoecetes bairdi*) reared in acidified waters (Walther *et al.*, 2010; Keppel *et al.*, 2012; Long *et al.*, 2013b), but intermoult duration decreased in pH 7.89 water and increased in pH 7.64 waters for adult shrimp (*Palaemon pacificus*) (Kurihara *et al.*, 2008). The percent increase in CL was not affected by acidified waters, but decreased in the warmest

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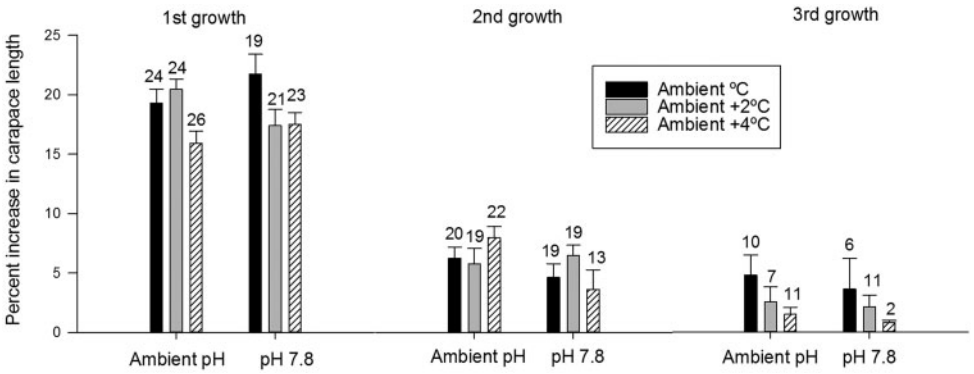

Figure 5. Mean young-of-the-year percent increase in CL for all treatments and each period of growth. Bars are mean ± s.e. and numbers above bars are sample sizes.

Table 5. General linear models of young-of-the-year red king crab size throughout the experiment.

	Factor	*F*-ratio	*p*-value
Initial CL	pH	1.574	0.212
	Temp	2.746	0.067
	pH × Temp	0.152	0.859
CL 1	pH	0.427	0.514
	Temp	1.422	0.245
	pH × Temp	0.441	0.644
CL 2	pH	2.651	0.106
	Temp	0.685	0.506
	pH × Temp	0.771	0.465
CL 3	pH	0.306	0.583
	Temp	1.412	0.255
	pH × Temp	0.301	0.742
Final CL	pH	0.013	0.910
	Temp	0.292	0.749
	pH × Temp	0.479	0.623

"CL" denotes carapace length, "pH" denotes treatment pH, "Temp" denotes treatment temperature and "pH × Temp" denotes the model interaction term.

Table 6. Fully crossed general linear model of young-of-the-year red king crab percent increase in size.

Factor	*F*-ratio	*p*-value
pH	2.248	0.136
Temp	3.125	0.047
G	282.142	<0.001
pH × Temp	0.377	0.687
pH × G	1.092	0.339
Temp × G	1.775	0.138
pH × Temp × G	1.832	0.126
Crab (pH × Temp)	1.601	0.003

"pH" denotes treatment pH, "Temp" denotes treatment temperature, "G" denotes periods of growth number; e.g. between initial and first moult, first moult and second moult, etc., and "Crab" denotes individual crab that were nested in the model.

temperatures in this study. Among early stage juvenile European lobster, elevated temperature increased inter-moult growth (percent change in wet body mass), but pH 7.74 waters did not, nor was there an interaction between elevated pCO_2 and temperature on inter-moult growth (Small *et al.*, 2016). The effect of increasing temperature on crustacean growth increment is variable, with the most common response being a decrease in growth increments (Hartnoll, 2001) as was observed in this study. Last, mean CL did not vary with treatment throughout or at the end of the experiment. This result is in contrast to another juvenile red king crab growth study in which larger crab were observed in warmer waters (Stoner *et al.*, 2010). The different results between the two experiments may be due to increased mortality with temperature and low sample sizes at the end of our experiment. Similar to this study, shrimp *P. pacificus* growth was not affected in pH 7.89 waters (Kurihara *et al.*, 2008), southern Tanner crab, *C. bairdi*, juvenile growth was not affected in pH 7.8 waters (Long *et al.*, 2013b), and there were no clear effects of acidified waters (pHs 7.82, 7.62) on European lobster larvae growth (Agnalt *et al.*, 2013). However, some decapod growth is affected by moderate levels of acidification. For example, American lobster larval growth was reduced in pH 7.7 waters at each stage (Keppel *et al.*, 2012).

In this study, pH 7.8 water did not substantially affect young-of-the-year red king crab growth, which is in contrast to the reduced growth observed in another young-of-the-year red king crab study with crab exposed to pH 7.8 water for approximately the same duration (Long *et al.*, 2013b). The Long *et al.* (2013b) experiment began with smaller and presumably younger crab than this study. Earlier juvenile instars may be more sensitive to acidified waters, thus the difference in responses between the two experiments. There can be considerable variability in sensitivity to acidification among life history stages within a species; southern Tanner crab (*C. bairdi*) and the porcelain crab both have increased mortality under acidified conditions at the juvenile stage, but not at the larval stage (Ceballos-Osuna *et al.*, 2013; Long *et al.*, 2013b, 2016) and megalops of *H. araneus* and *H. gammarus* are sensitive to acidification, but other larval stages are not (Walther *et al.*, 2010; Small *et al.*, 2015). Alternately, ocean acidification studies on early life-history stages of the porcelain crab suggest that responses are brood-specific and possibly the result of genetic variation, environmental factors, and maternal condition during egg production (Carter *et al.*, 2013; Ceballos-Osuna *et al.*, 2013; Stillman and Paganini, 2015). The differences observed between the two young-of-the-year red king crab studies may also have been due to these factors, since juveniles came from different mothers from different micro-environments.

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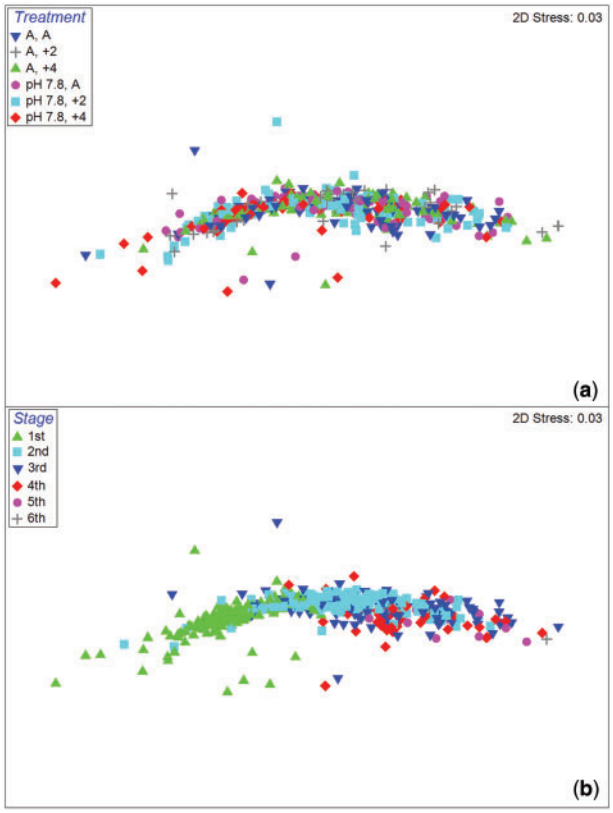

Figure 6. n-MDS plots of young-of-the-year red king crab carapace morphometrics showing variations among (a) treatments and (b) moult stages. Treatments are indicated in the key as pH treatment, temperature treatment. "A" represents ambient for both pH and temperature, +2 and +4 represent the ambient +2 °C and ambient +4 °C temperature treatments. Stage represents moult stage with first representing the initial, second the moult stage after the first moult, and so on.

Maternal size does not affect red king crab embryo or larval quality (Swiney *et al.*, 2013a), but there is variability in quality among broods. Regardless of the exact mechanism, the variation in young-of-the-year red king crab growth responses to ocean acidification suggests that this life history stage has the potential to adapt to some levels of ocean acidification, but one should not forget that mortality rates increased with acidification in both studies. Carter *et al.* (2013) suggest that to be more informative when predicting effects of ocean acidification across populations, it is important to look at variations in responses across multiple broods rather than focusing on mean responses.

Young-of-the-year red king crab morphology did not vary with temperature or pH in this study. Likewise, young-of-the-year red king crab morphology did not vary under acidified conditions in another study (Long *et al.*, 2013b), nor were abnormalities in the Norway lobster *Nephrops norvegicus* embryos observed under elevated temperature or acidified conditions (Styf *et al.*, 2013). Acidified waters do alter morphology in other decapods. The shrimp *P. pacificus* had shorter second antenna in acidified waters, which may alter the shrimp's behavior since this antenna provides most of the tactile information they use to explore the environment (Kurihara *et al.*, 2008). Few studies have looked at ocean acidification effects on decapod juvenile morphology, with the exception that no effects were found with Tanner crab juveniles (Long *et al.*, 2013b), but European lobster juveniles had deformities that may affect respiration, the ability to find food or mates, and the ability to swim (Agnalt *et al.*, 2013).

Table 7. PERMANOVA model of young-of-the-year red king crab morphology with temperature (T), pH, and stage (S) as factors.

Factor	df	Pseudo-F	*p*-value
Temperature	2	1.466	0.197
pH	1	1.417	0.214
Stage	5	109.180	0.001
T × pH	2	0.528	0.627
T × S	8	0.333	0.965
pH × S	4	0.108	0.993
T × pH × S	7	0.922	0.504

In general, crustaceans are not as sensitive to ocean acidification and warming temperature as other taxa in the short to mid duration that most meta-data analysis are based on (Byrne and Przeslawski, 2013; Harvey *et al.*, 2013; Kroeker *et al.*, 2013), but as this and other studies illustrate, red king crab are affected by these stressors. Of the stages and parameters studied to date, the most potentially harmful effects of ocean acidification on red king crab populations are seen in low survival rates among young-of-the-year red king crab, with 100% mortality observed after 95 d in pH 7.5 water (Long *et al.*, 2013b). At the end of this experiment, only 3% of the crabs survived in the pH 7.8 ambient +4 °C temperature treatment. If these low survival rates are observed *in situ* under predicted acidification and warming conditions, recruitment to the fishery would likely not be sufficient to allow the current harvest levels, or potentially any harvest at all (Punt *et al.*, 2014). Carryover effects of ocean acidification between stages occur in southern Tanner crab; in particular, oogenesis appears to be particularly sensitive to ocean acidification and affects carryover and altered embryonic development, with reduced hatching success, while larvae were smaller, with lower calcium and magnesium contents and lower metabolic rates (Long *et al.*, 2016; Swiney *et al.*, 2016). Given that both the embryos and larvae of red king crab are sensitive to ocean acidification (Long *et al.*, 2013a), it is feasible, if not likely, that similar responses would be seen if red king crab are exposed to acidified waters for their entire reproductive cycle. The results of this study combined with other studies suggest that ocean acidification and warming may have profound negative effects on red king crab populations in the upcoming decades unless the species is able to quickly adapt or acclimate to changing conditions. Stock assessment and management of red king crab will likely be improved if the effects of ocean acidification on the populations are considered.

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